

Computer Networks

Lecture 12: Data Link Layer

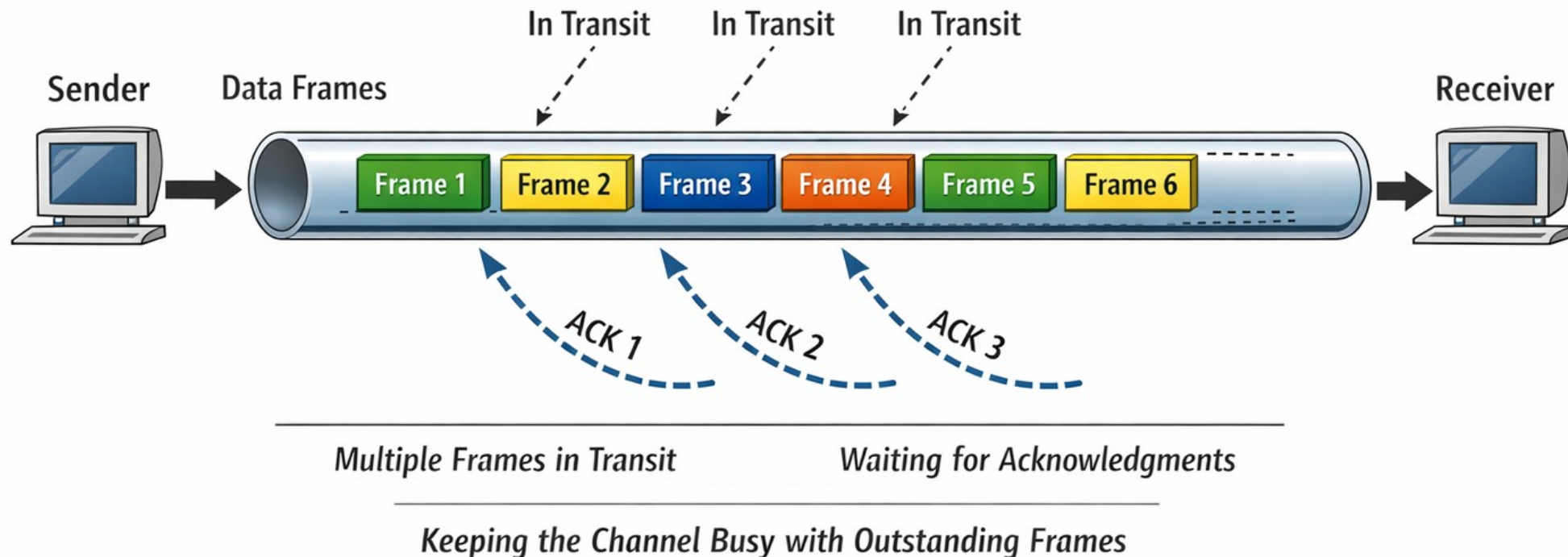
Content



- **Go-Back-N Automatic Repeat Request**

Go-Back-N Automatic Repeat Request

To improve the efficiency of transmission multiple frames must be in transition while waiting for acknowledgment. In other words, we need to let more than one frame be outstanding to keep the channel busy while the sender is waiting for acknowledgment



Go-Back-N Automatic Repeat Request

Sequence Number

If the header of the frame allows m bits for the sequence number, the sequence numbers range from 0 to $2^m - 1$.

Example

If m is 4, the only sequence numbers are 0 through 15 inclusive. However, we can repeat the sequence. So the sequence numbers are

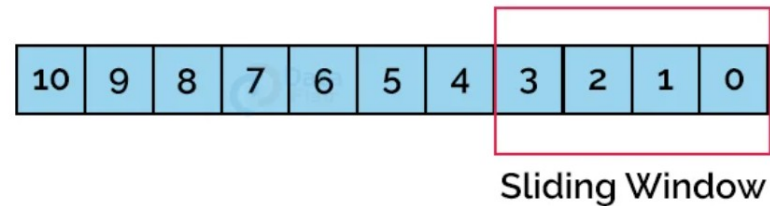
0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,0,1

Go-Back-N Automatic Repeat Request

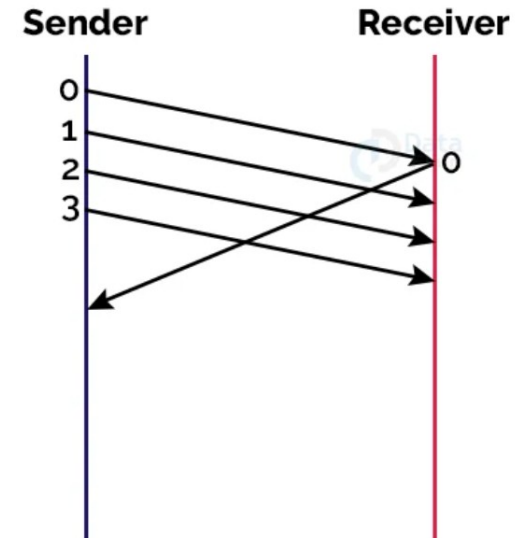
Sliding Window

It allows the sender to send multiple frames before waiting for acknowledgments (ACKs).

- The range which is the concern of the sender is called the send sliding window.
- The range that is the concern of the receiver is called the receive sliding window.



Window Size : 4



Go-Back-N Automatic Repeat Request

Send Window

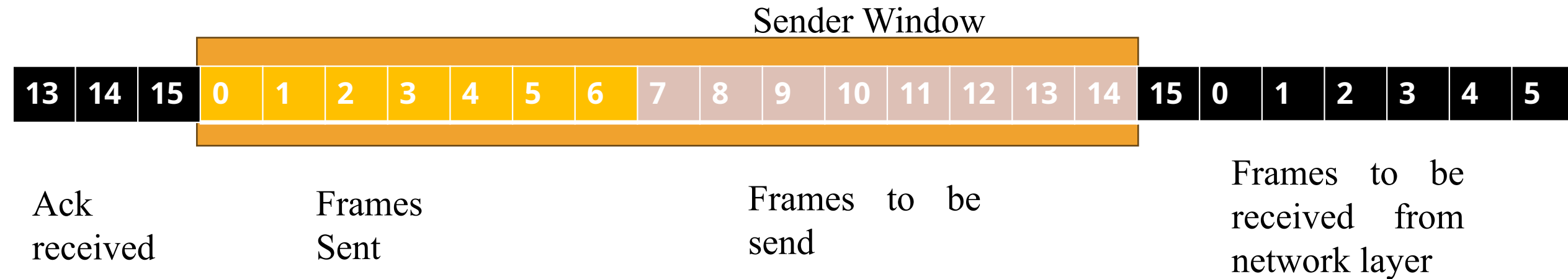
The send window is a logical (imaginary) range of sequence numbers. It represents which frames are allowed to be in transit

- The window covers a set of sequence numbers of data frames.
- Only frames whose sequence numbers lie inside the window can be transmitted.

The maximum size of the window is $2^m - 1$

Go-Back-N Automatic Repeat Request

Send Window



In Go-Back-N, the sender window is controlled using three variables:

Sf = Sequence number of the first outstanding (oldest unacknowledged) frame

Sn = Sequence number of the next frame to be sent

Ssize = Maximum number of frames allowed to be outstanding

Go-Back-N Automatic Repeat Request

Receive Window

The receive window makes sure that the correct data frames are received and that the correct acknowledgments are sent. The receiver is always looking for the arrival of a specific frame.

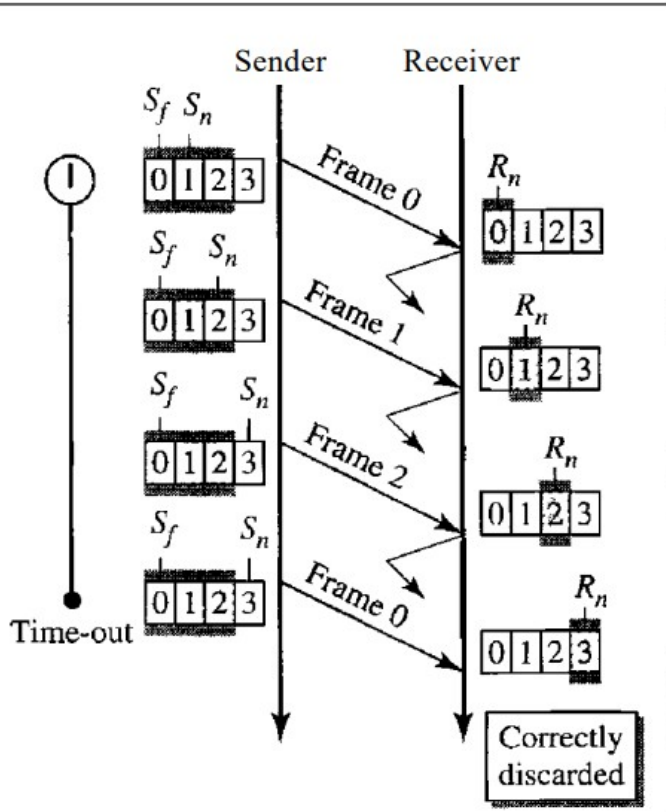
Any frame arriving out of order is discarded and needs to be resent

Sender

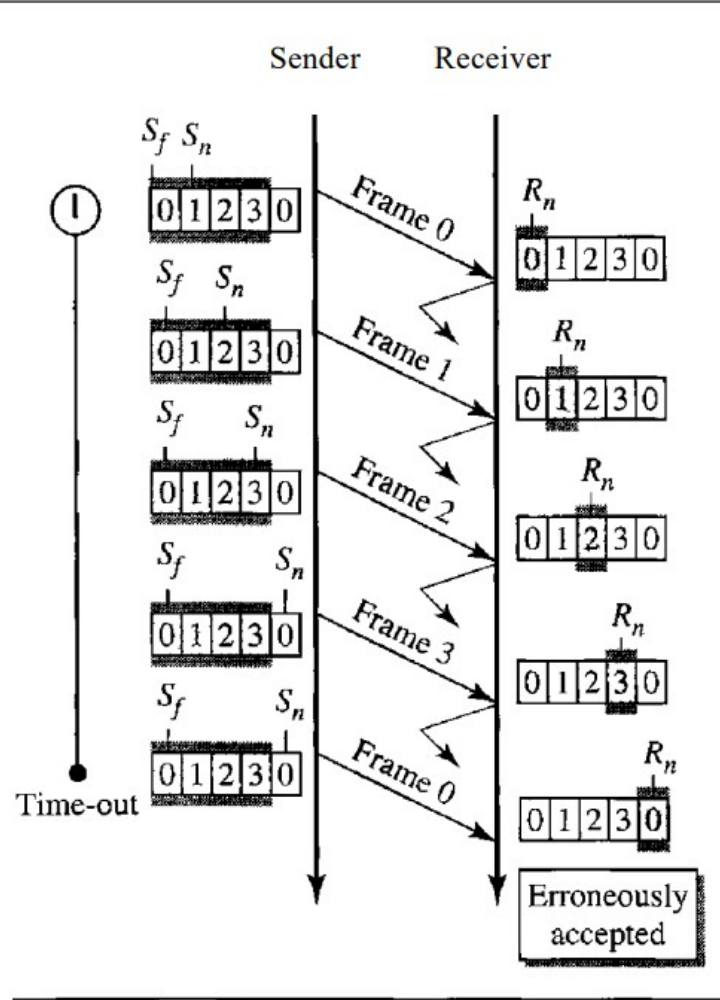
Receiver



The size of the receive window is always 1.



a. Window size $< 2^m$



b. Window size $= 2^m$

Go-Back-N Automatic Repeat Request

Why Send window size is $2^n - 1$

Questions

1. In *go back N ARQ* if sequence bits are taken as 5 then what is the sender window size and receiver window size.
Ans: 31
2. In *go back N ARQ* if sequence bits are taken as 6 then what is the sender window size and receiver window size.
Ans: 63
3. In M is the max sequence no. in sliding window of *go back N ARQ* . Calculate the no. of frames that can be transmitted in RTT.
Ans: M
4. What will be the sequence no. of 30th frame if 3 bit sequence no. is used in *go back N ARQ* .
Ans: 5
5. If 5 bit sequence no. is used what will be the sequence no. after sending 100 frames.
Ans: 4

Questions

1. If the size of the window is N and N no. of frames are transmitted. It is supporting cumulative acknowledgement. The ack no. that is transmitted by the receiver is P .
 - a) Calculate how many frames are received by the receiver
 - b) How many frames are to be retransmitted.

Ans: P

Ans: $N - P$

2. If K is the sequence bits that are used and L is the cumulative ack
 - a) How many frames have reached to the destination
 - b) How many frames are to be retransmitted

Ans: L

Ans: $2^K - 1 - L$

3. If sender window size in *GO back N ARQ* is Q . Calculate the no. of bits taken.

Ans: $\log_2(Q + 1)$

Questions

4. *Bandwidth = 10Mbps, RTT = 10 μ sec, Frame size = 50 bits, Go back NARQ is used. Calculate window size, no. of sequence bits.*

$$1 \text{ sec} \rightarrow 10^7 \text{ bits}$$

$$10\mu\text{sec} \rightarrow 100 \text{ bits}$$

$$\text{No. of frames} = 100/50 = 20$$

$$\text{Window size} = 2$$

$$\text{No. of sequence bits} = 2$$